

Invariant Subspaces and Irreducibility

Road map

Diagonalisation is one of the central problems of ordinary matrix theory. For a family of matrices representing a group, the analogous problem is to find one basis, valid for every group element, in which all matrices have the smallest possible blocks. The correct language is that of invariant subspaces, reducible and irreducible representations, and complete reducibility. After developing this language, we prove the orthogonal complement theorem, Maschke's theorem, Schur's lemma and its principal consequences, the one-dimensionality of complex irreducible representations of Abelian groups, and uniqueness of irreducible multiplicities.

22.1 Invariant subspaces and restricted representations

Let

$$D : G \longrightarrow GL(V)$$

be a representation of a group G on a vector space V .

Definition 22.1: G -invariant subspace

A linear subspace $U \leq V$ is *invariant under G* , or *D -invariant*, if

$$D(a)U \subseteq U \quad \text{for every } a \in G. \quad (22.1)$$

Equivalently, $u \in U$ implies $D(a)u \in U$ for every $a \in G$.

Because $D(a)$ is invertible and $D(a)^{-1} = D(a^{-1})$, the inclusion in Equation (22.1) is actually an equality:

$$D(a)U = U.$$

Indeed, $D(a)U \subseteq U$, while applying invariance to a^{-1} gives $D(a^{-1})U \subseteq U$; multiplication by $D(a)$ then gives $U \subseteq D(a)U$.

The two *trivial invariant subspaces* are always

$$\{0\} \quad \text{and} \quad V.$$

Any invariant subspace different from both is called nontrivial.

If U is invariant, every $D(a)$ restricts to an invertible linear map on U . We may therefore define

$$D_U : G \longrightarrow GL(U), \quad D_U(a) := D(a)|_U. \quad (22.2)$$

For $u \in U$,

$$D_U(a)D_U(b)u = D(a)D(b)u = D(ab)u = D_U(ab)u,$$

so D_U is a representation. If U is nontrivial and $\dim V < \infty$, then

$$0 < \dim D_U = \dim U < \dim V = \dim D.$$

This is the first way in which an invariant subspace reduces the effective dimension of a representation.

Proposition 22.1: Restriction of a unitary representation

If V is an inner-product space, D is unitary and U is invariant, then D_U is unitary with respect to the inner product restricted to U .

证明. For $u_1, u_2 \in U$ and $a \in G$,

$$\langle D_U(a)u_1, D_U(a)u_2 \rangle = \langle D(a)u_1, D(a)u_2 \rangle = \langle u_1, u_2 \rangle.$$

Thus each $D_U(a)$ preserves the restricted inner product. ■

22.2 Reducibility and block-triangular form

Definition 22.2: Reducible and irreducible representations

The representation $D : G \rightarrow GL(V)$ is *reducible* if V contains a nontrivial invariant subspace:

$$\exists U \leq V, \quad U \neq \{0\}, \quad U \neq V, \quad D(a)U \subseteq U \quad (a \in G).$$

If no such subspace exists, D is *irreducible*.

22.2.1 An adapted basis

Suppose D is reducible and U is an invariant subspace with

$$\dim U = k, \quad \dim V = n, \quad 0 < k < n.$$

Choose a basis $(e_1, \dots, e_k)$ of U and extend it to a basis

$$\mathcal{B} = (e_1, \dots, e_k, e_{k+1}, \dots, e_n)$$

of V . Define

$$W := \text{span}\{e_{k+1}, \dots, e_n\}.$$

As vector spaces,

$$V = U \oplus W,$$

but W has not been assumed invariant.

Write

$$D(a)e_j = \sum_{i=1}^n D_{ij}(a)e_i.$$

For $1 \leq j \leq k$, invariance gives $D(a)e_j \in U$, so the coefficients of $e_{k+1}, \dots, e_n$ vanish:

$$D_{ij}(a) = 0 \quad (1 \leq j \leq k, k+1 \leq i \leq n).$$

Consequently every representing matrix has the same block-upper-triangular form

$$[D(a)]_{\mathcal{B}} = \begin{pmatrix} D_U(a) & N(a) \\ 0 & D''(a) \end{pmatrix}. \quad (22.3)$$

Here the lower-left zero block expresses invariance of U . The block $N(a)$ measures how vectors chosen in W may acquire components in U .

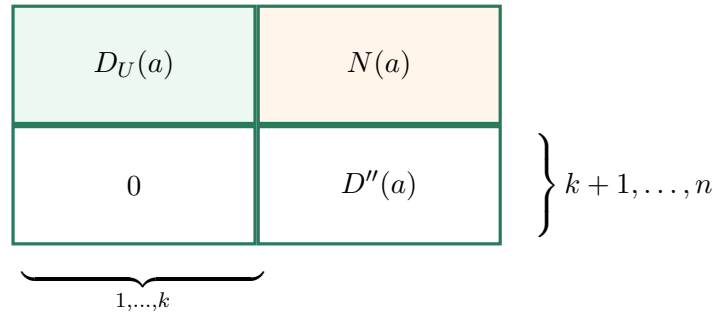


图 22.1: A reducible representation in a basis adapted to one invariant subspace. Only the lower-left block is forced to vanish.

If $u \in U$, its adapted coordinate column has the form

$$[u]_{\mathcal{B}} = \begin{pmatrix} x \\ 0 \end{pmatrix},$$

and Equation (22.3) gives

$$[D(a)u]_{\mathcal{B}} = \begin{pmatrix} D_U(a)x \\ 0 \end{pmatrix} \in U.$$

In contrast, for $w \in W$,

$$[w]_{\mathcal{B}} = \begin{pmatrix} 0 \\ y \end{pmatrix} \implies [D(a)w]_{\mathcal{B}} = \begin{pmatrix} N(a)y \\ D''(a)y \end{pmatrix}.$$

The first component need not vanish; thus $D(a)w \notin W$ in general.

22.2.2 Example: rotations about one axis

Let

$$G = \{R_z(\theta) : \theta \in \mathbb{R}\} \leq SO(3)$$

act on $V = \mathbb{R}^3$ by the defining representation $D(R) = R$. The xy -plane

$$U = \text{span}\{e_x, e_y\}$$

and the z -axis

$$\bar{U} = \text{span}\{e_z\}$$

are both invariant. If W is a slanted line complementary to U , however, rotating its nonzero horizontal component takes it out of W .

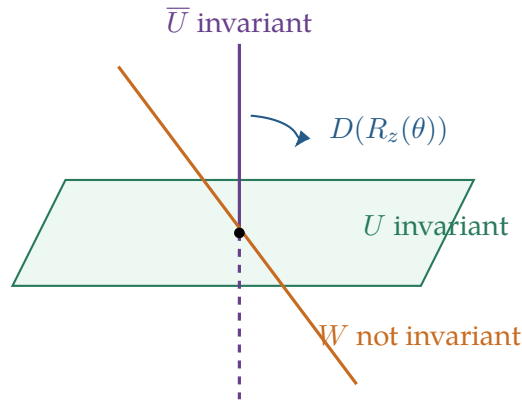


图 22.2: Rotations about the z -axis preserve the horizontal plane and the z -axis, but not a generic complementary line.

Thus the three-dimensional representation D is reducible. Its restriction to U is the usual real two-dimensional rotation representation, and its restriction to $\bar{U}$ is the one-dimensional trivial representation. The restriction D_W cannot be defined because W is not invariant. Over $\mathbb{R}$, the action of the full rotation circle on U is irreducible: a real invariant line would have to be preserved by every rotation, which is impossible. Over $\mathbb{C}$, its complexification splits into two one-dimensional weight spaces; thus the ground field matters.

22.3 Completely reducible representations

If both U and a complementary subspace W are invariant, then $N(a) = 0$ in Equation (22.3), and

$$V = U \oplus W, \quad D = D_U \oplus D_W.$$

In an adapted basis,

$$[D(a)]_{\mathcal{B}} = \begin{pmatrix} D_U(a) & 0 \\ 0 & D_W(a) \end{pmatrix} \quad (a \in G). \quad (22.4)$$

One-dimensional representations

Every one-dimensional representation is irreducible, because a one-dimensional vector space has no subspaces other than $\{0\}$ and itself.

Definition 22.3: Completely reducible representation

A representation $D : G \rightarrow GL(V)$ is *completely reducible* if its representation space is a direct sum of invariant irreducible subspaces,

$$V = \bigoplus_{\alpha} U_{\alpha},$$

and hence

$$D = \bigoplus_{\alpha} D_{\alpha}, \quad D_{\alpha} := D|_{U_{\alpha}}, \quad (22.5)$$

with every D_{α} irreducible.

For finite-dimensional V , complete reducibility is equivalent to the existence of one basis in which every $D(a)$ has the same block-diagonal pattern,

$$[D(a)] = \begin{pmatrix} D_1(a) & & 0 \\ & \ddots & \\ 0 & & D_r(a) \end{pmatrix},$$

where every block representation D_{α} is irreducible. The matrices inside the blocks vary with a , but the block sizes and positions do not.

22.3.1 A reducible representation that is not completely reducible

Consider the additive group $G = (\mathbb{R}, +)$ and

$$D(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}. \quad (22.6)$$

This is a real two-dimensional representation because

$$D(x)D(x') = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+x' \\ 0 & 1 \end{pmatrix} = D(x+x'),$$

and $D(0) = I$. Its determinant is 1, so every $D(x)$ is invertible.

The line

$$U = \left\{ \begin{pmatrix} u \\ 0 \end{pmatrix} : u \in \mathbb{R} \right\} = \text{span}\{e_1\}$$

is invariant, since

$$D(x) \begin{pmatrix} u \\ 0 \end{pmatrix} = \begin{pmatrix} u \\ 0 \end{pmatrix}.$$

Thus D is reducible.

Suppose it were completely reducible. Then there would be an invariant one-dimensional complement U' to U . Write

$$U' = \text{span} \left\{ \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} \right\}.$$

The condition $U' \neq U$ forces $u'_2 \neq 0$. But

$$D(x) \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = \begin{pmatrix} u'_1 + xu'_2 \\ u'_2 \end{pmatrix}.$$

If this vector were in U' , there would be a scalar λ_x such that

$$\begin{pmatrix} u'_1 + xu'_2 \\ u'_2 \end{pmatrix} = \lambda_x \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix}.$$

The second coordinate and $u'_2 \neq 0$ imply $\lambda_x = 1$; the first coordinate then implies $xu'_2 = 0$, which fails whenever $x \neq 0$. Therefore no invariant complement exists, and D is not completely reducible.

The exception and the rule

Representations need not be completely reducible, as the unipotent example shows. Nevertheless complete reducibility is the rule in several central settings: finite-dimensional unitary representations, finite-group representations over characteristic zero, and continuous finite-dimensional representations of compact groups.

22.4 Unitary representations and invariant complements

Theorem 22.1: Orthogonal complements are invariant

Let V be a finite-dimensional inner-product space and let

$$D : G \longrightarrow U(V)$$

be a unitary representation. If $U \leq V$ is invariant, then its orthogonal complement

$$U^\perp = \{w \in V : \langle u, w \rangle = 0 \text{ for every } u \in U\}$$

is invariant, and

$$V = U \oplus U^\perp, \quad D = D_U \oplus D_{U^\perp}.$$

证明. The set $U^\perp$ is a subspace. For instance, if $w_1, w_2 \in U^\perp$ and α, β are scalars, then for every $u \in U$,

$$\langle u, \alpha w_1 + \beta w_2 \rangle = \alpha \langle u, w_1 \rangle + \beta \langle u, w_2 \rangle = 0.$$

Choose an orthonormal basis

$$(e_1, \dots, e_k)$$

of U , and extend it to the orthonormal basis

$$(e_1, \dots, e_n)$$

of V . Then

$$U^\perp = \text{span}\{e_{k+1}, \dots, e_n\},$$

so every $v \in V$ has a unique orthogonal decomposition

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

Hence $V = U \oplus U^\perp$.

It remains to prove invariance. Let $w \in U^\perp$, $u \in U$, and $a \in G$. Because $D(a)$ is unitary,

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

Therefore

$$\begin{aligned} \langle u, D(a)w \rangle &= \langle D(a)^\dagger u, w \rangle \\ &= \langle D(a^{-1})u, w \rangle. \end{aligned}$$

Since U is invariant, $D(a^{-1})u \in U$; since $w \in U^\perp$, the last inner product is zero. Thus

$$\langle u, D(a)w \rangle = 0 \quad \text{for every } u \in U,$$

which says $D(a)w \in U^\perp$. Hence $U^\perp$ is invariant. ■

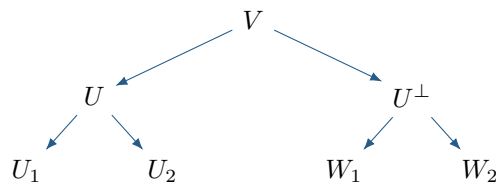
Theorem 22.2: Finite-dimensional unitary representations are completely reducible

Every finite-dimensional unitary representation is completely reducible.

证明. If D is irreducible, there is nothing to prove. Otherwise choose a nontrivial invariant subspace U . By the orthogonal-complement theorem above,

$$V = U \oplus U^\perp, \quad D = D_U \oplus D_{U^\perp}.$$

If either restricted representation is reducible, split that invariant subspace again into a nontrivial invariant subspace and its orthogonal complement. Continue this procedure:



At every genuine split, each child has strictly smaller positive dimension than its parent. Since $\dim V < \infty$, the process must terminate after finitely many steps. The terminal invariant subspaces are irreducible and their orthogonal direct sum is V . Hence D is completely reducible. ■

22.5 Maschke's theorem

Chapter 21 showed that a finite-group representation over $\mathbb{C}$ is equivalent to a unitary representation. We can now combine that result with the complete-reducibility theorem for unitary representations.

Theorem 22.3: Maschke's theorem over $\mathbb{R}$ or $\mathbb{C}$

Let G be finite and let $D : G \rightarrow GL(V)$ be a finite-dimensional real or complex representation. Then D is completely reducible.

Proof by unitarisation and pullback. Average an inner product over G as in Equation (21.5). With respect to this averaged inner product, D is unitary. Hence

$$V = U'_1 \oplus \cdots \oplus U'_r, \quad D' = D'_1 \oplus \cdots \oplus D'_r,$$

for irreducible invariant subspaces U'_α , after identifying D with an equivalent unitary representation D' . Let

$$D(a) = S^{-1} D'(a) S$$

and define

$$U_\alpha := S^{-1} U'_\alpha.$$

First, $V = \bigoplus_\alpha U_\alpha$. For any $v \in V$, write

$$Sv = u'_1 + \cdots + u'_r, \quad u'_\alpha \in U'_\alpha.$$

Then

$$v = S^{-1} u'_1 + \cdots + S^{-1} u'_r,$$

so the U_α span V . If

$$u_1 + \cdots + u_r = \bar{u}_1 + \cdots + \bar{u}_r, \quad u_\alpha, \bar{u}_\alpha \in U_\alpha,$$

applying S gives two decompositions in $\bigoplus_\alpha U'_\alpha$, hence $Su_\alpha = S\bar{u}_\alpha$ and, because S is injective, $u_\alpha = \bar{u}_\alpha$.

Second, each U_α is invariant:

$$D(a)U_\alpha = S^{-1} D'(a) S S^{-1} U'_\alpha = S^{-1} D'(a) U'_\alpha \subseteq S^{-1} U'_\alpha = U_\alpha.$$

Finally, $D|_{U_\alpha}$ is irreducible. If $\bar{U}_\alpha$ were a nontrivial invariant subspace of U_α , then $S\bar{U}_\alpha$ would be a nontrivial invariant subspace of U'_α , contradicting irreducibility of D'_α . Therefore the pulled-back decomposition is a complete reduction of D . ■

22.5.1 The averaged-projection proof

There is a more algebraic proof that displays directly how to manufacture an invariant complement. Let $U \leq V$ be invariant. Choose any vector-space complement W , so $V = U \oplus W$, and let

$$P : V \longrightarrow U$$

be the projection along W : $P(u + w) = u$. The map P need not commute with D . Average all its conjugates:

$$\bar{P} := \frac{1}{|G|} \sum_{a \in G} D(a) P D(a)^{-1}. \quad (22.7)$$

For $b \in G$,

$$\begin{aligned} D(b)\overline{P}D(b)^{-1} &= \frac{1}{|G|} \sum_{a \in G} D(ba)PD(ba)^{-1} \\ &= \overline{P}, \end{aligned}$$

because $a \mapsto ba$ permutes G . Hence

$$D(b)\overline{P} = \overline{P}D(b).$$

Moreover, for $u \in U$, invariance gives $D(a)^{-1}u \in U$, so

$$D(a)PD(a)^{-1}u = D(a)D(a)^{-1}u = u.$$

Thus $\overline{P}|_U = I_U$, while every summand in Equation (22.7) has image in U ; consequently $\text{im } \overline{P} = U$ and $\overline{P}^2 = \overline{P}$. Its kernel is an invariant complement:

$$V = U \oplus \ker \overline{P}.$$

Indeed, commutation gives

$$\overline{P}(D(b)v) = D(b)\overline{P}(v) = 0 \quad (v \in \ker \overline{P}).$$

Repeatedly applying this construction proves complete reducibility.

General field and compact-group forms

The projection proof works over a field $\mathbb{K}$ whenever $|G|$ is invertible in $\mathbb{K}$, that is, $\text{char } \mathbb{K} \nmid |G|$. This is the usual algebraic form of Maschke's theorem. If the characteristic divides $|G|$, complete reducibility can fail.

For a compact group, the finite average is replaced by normalized Haar integration:

$$\overline{P} = \int_G D(a)PD(a)^{-1} d\mu(a).$$

Thus every continuous finite-dimensional representation of a compact group over $\mathbb{R}$ or $\mathbb{C}$ is completely reducible.

22.5.2 A comparison of the hypotheses

The preceding results are easy to confuse, so it is useful to display their logical roles explicitly.

Hypothesis or ingredient	Invariant orthogonal complement	Unitary complete reduction	Maschke / compact averaging
$ G < \infty$	—	—	sufficient
Normalized group average exists	—	—	essential
$\dim V < \infty$	used for $V = U \oplus U^\perp$ here	ensures termination	standard assumption
V has an inner product	required	required	constructed if needed
D is unitary	required	required	obtained by averaging

表 22.1: How the hypotheses enter the complete-reducibility theorems.

22.6 Schur's lemma

A central theorem of representation theory is Schur's lemma—so central that it is often simply called *the lemma of representation theory*. Let

$$D : G \rightarrow GL(V), \quad D' : G \rightarrow GL(V')$$

be irreducible representations, and let $A : V \rightarrow V'$ be an intertwining map:

$$D'(a)A = AD(a) \quad (a \in G). \quad (22.8)$$

The equation is represented by the commutative square

$$\begin{array}{ccc}
 V & \xrightarrow{A} & V' \\
 D(a) \downarrow & & \downarrow D'(a) \\
 V & \xrightarrow{A} & V'
 \end{array}$$

Theorem 22.4: Schur's lemma, intertwiner form

For irreducible representations D on V and D' on V' , an intertwiner $A : V \rightarrow V'$ satisfies exactly one of the following:

1. $A = 0$;
2. A is an isomorphism, so $D \sim D'$ and $\dim V = \dim V'$.

No finite-dimensional hypothesis is needed for this algebraic statement.

证明. We first show that $\ker A \leq V$ is D -invariant. If $v \in \ker A$, then $Av = 0$, and

$$A(D(a)v) = D'(a)Av = 0.$$

Thus $D(a)v \in \ker A$. Irreducibility of D implies

$$\ker A = \{0\} \quad \text{or} \quad \ker A = V.$$

Next, $\text{im } A \leq V'$ is D' -invariant. If $v' \in \text{im } A$, then $v' = Av$ for some $v \in V$, and

$$D'(a)v' = D'(a)Av = AD(a)v \in \text{im } A.$$

Irreducibility of D' implies

$$\text{im } A = \{0\} \quad \text{or} \quad \text{im } A = V'.$$

If $\ker A = V$, then $A = 0$. If $\text{im } A = \{0\}$, again $A = 0$. These are the first alternative. Otherwise $\ker A = \{0\}$ and $\text{im } A = V'$, so A is both injective and surjective and hence is an isomorphism. Rearranging the intertwining relation gives

$$D'(a) = AD(a)A^{-1},$$

so $D' \sim D$. This is the second alternative. ■

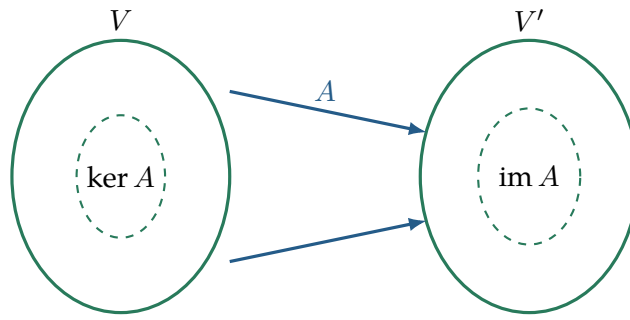


图 22.3: The kernel and image of an intertwiner are invariant. For irreducible source and target they can only be trivial or the whole space.

Proposition 22.2: Scalar commutant over $\mathbb{C}$

Let $D : G \rightarrow GL(V)$ be an irreducible representation on a nonzero finite-dimensional complex vector space. If $A : V \rightarrow V$ commutes with the representation,

$$[D(a), A] = D(a)A - AD(a) = 0 \quad (a \in G),$$

then

$$A = \lambda I_V \quad \text{for some } \lambda \in \mathbb{C}.$$

证明. Because $V \neq \{0\}$ is finite-dimensional over $\mathbb{C}$, the characteristic polynomial of A has a root. Let λ be an eigenvalue and choose $v \neq 0$ with $Av = \lambda v$. Define

$$\bar{A} := A - \lambda I_V.$$

The map $\bar{A}$ also commutes with every $D(a)$, so it is an intertwiner from D to itself. Moreover,

$$\bar{A}v = (A - \lambda I_V)v = 0,$$

and hence $\ker \bar{A} \neq \{0\}$. By Schur's intertwiner lemma above, a nonzero intertwiner of an irreducible representation with itself must be an isomorphism. Since $\bar{A}$ has nonzero kernel it is not an isomorphism, so the only possibility is $\bar{A} = 0$. Therefore $A = \lambda I_V$. ■

Why the complex field matters

The eigenvalue step is automatic over $\mathbb{C}$ but not over $\mathbb{R}$. A real irreducible representation can have a larger division algebra of commuting operators; real, complex and quaternionic types all occur. The scalar conclusion above should therefore not be transferred to real representations without additional hypotheses.

22.7 The commutant and inequivalent irreducible summands

Suppose a complex representation decomposes as

$$D = D_1 \oplus \cdots \oplus D_r$$

on

$$V = V_1 \oplus \cdots \oplus V_r,$$

where the D_α are irreducible and pairwise inequivalent. Let $A : V \rightarrow V$ commute with D .

For clarity, first take $r = 2$. In the corresponding block basis write

$$D(a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}, \quad A = \begin{pmatrix} A_1 & Q_1 \\ Q_2 & A_2 \end{pmatrix}.$$

The equation $D(a)A = AD(a)$ becomes

$$\begin{pmatrix} D_1(a)A_1 & D_1(a)Q_1 \\ D_2(a)Q_2 & D_2(a)A_2 \end{pmatrix} = \begin{pmatrix} A_1D_1(a) & Q_1D_2(a) \\ Q_2D_1(a) & A_2D_2(a) \end{pmatrix}.$$

Thus

$$\begin{aligned} D_1(a)A_1 &= A_1D_1(a), & D_2(a)A_2 &= A_2D_2(a), \\ D_1(a)Q_1 &= Q_1D_2(a), & D_2(a)Q_2 &= Q_2D_1(a). \end{aligned}$$

The scalar-commutant form of Schur's lemma gives

$$A_1 = \lambda_1 I_{V_1}, \quad A_2 = \lambda_2 I_{V_2}.$$

The off-diagonal block $Q_1 : V_2 \rightarrow V_1$ is an intertwiner from D_2 to D_1 . Since these irreducibles are inequivalent, Schur's intertwiner lemma forces $Q_1 = 0$. Similarly $Q_2 = 0$. Therefore

$$A = \begin{pmatrix} \lambda_1 I_{V_1} & 0 \\ 0 & \lambda_2 I_{V_2} \end{pmatrix}.$$

The same block-by-block argument for arbitrary r gives

$$A = \lambda_1 I_{V_1} \oplus \cdots \oplus \lambda_r I_{V_r}. \quad (22.9)$$

Thus each irreducible invariant subspace V_α is a subspace on which every commuting operator acts by a scalar. If the numbers λ_α are pairwise distinct, the V_α are exactly the eigenspaces of A . If some scalars coincide, the corresponding eigenspace is the direct sum of all V_α carrying that common scalar. This qualification makes precise the eigenspace statement in the handwritten argument.

22.8 Irreducible representations of Abelian groups

Theorem 22.5: Complex irreducibles of Abelian groups

Every nonzero finite-dimensional irreducible complex representation of an Abelian group is one-dimensional.

证明. Let $D : G \rightarrow GL(V)$ be irreducible and let $a \in G$. Since G is Abelian, for every $b \in G$,

$$D(a)D(b) = D(ab) = D(ba) = D(b)D(a).$$

Thus $D(a)$ commutes with the whole representation. By the scalar-commutant form of Schur's lemma,

$$D(a) = \lambda(a)I_V$$

for some nonzero $\lambda(a) \in \mathbb{C}$. Choose any nonzero $e \in V$. The line $L = \text{span}\{e\}$ is invariant because

$$D(a)e = \lambda(a)e \in L \quad (a \in G).$$

Irreducibility and $L \neq \{0\}$ force $L = V$. Hence $\dim V = 1$. ■

Example 22.1: Simultaneous diagonalisation of an Abelian unitary action

Let $D : G \rightarrow U(V)$ be a finite-dimensional unitary representation of an Abelian group. Every $D(a)$ is normal, and all the $D(a)$ commute:

$$D(a)D(b) = D(ab) = D(ba) = D(b)D(a).$$

A commuting family of normal operators on a finite-dimensional complex inner-product space is simultaneously unitarily diagonalizable. Hence there is an orthonormal basis $(e_1, \dots, e_n)$ such that

$$D(a)e_\alpha = \lambda_\alpha(a)e_\alpha \quad (a \in G).$$

Each line

$$U_\alpha = \text{span}\{e_\alpha\}$$

is invariant and carries the one-dimensional representation

$$D_\alpha(a) = \lambda_\alpha(a).$$

Therefore

$$V = U_1 \oplus \dots \oplus U_n, \quad D = D_1 \oplus \dots \oplus D_n.$$

This is the concrete matrix form of the theorem: every finite-dimensional unitary representation of an Abelian group decomposes into one-dimensional irreducibles.

22.9 Uniqueness of irreducible multiplicities

Theorem 22.6: Uniqueness of a complete reduction

Let D be a finite-dimensional completely reducible complex representation. Grouping equivalent summands, write

$$D \sim n_1 D^{(1)} \oplus n_2 D^{(2)} \oplus \cdots \oplus n_r D^{(r)}, \quad (22.10)$$

where the $D^{(\alpha)}$ are pairwise inequivalent irreducible representations and

$$n_\alpha D^{(\alpha)} := \underbrace{D^{(\alpha)} \oplus \cdots \oplus D^{(\alpha)}}_{n_\alpha \text{ copies}}.$$

Then the set of irreducible equivalence classes $\{D^{(1)}, \dots, D^{(r)}\}$ and every multiplicity n_α are uniquely determined by D , up to reordering.

证明. For representations E on W and D on V , let

$$\text{Hom}_G(E, D) = \{A : W \rightarrow V : AE(a) = D(a)A \text{ for every } a \in G\}$$

be the vector space of intertwiners.

Fix an irreducible representation E . If $D = \bigoplus_{\alpha,j} D^{(\alpha)}$, then an intertwiner $A : E \rightarrow D$ is equivalent to a collection of component intertwiners

$$A_{\alpha,j} : E \longrightarrow D^{(\alpha)}.$$

Hence

$$\text{Hom}_G(E, D) \cong \bigoplus_{\alpha=1}^r \bigoplus_{j=1}^{n_\alpha} \text{Hom}_G(E, D^{(\alpha)}).$$

Schur's lemma says

$$\dim_{\mathbb{C}} \text{Hom}_G(E, D^{(\alpha)}) = \begin{cases} 1, & E \sim D^{(\alpha)}, \\ 0, & E \not\sim D^{(\alpha)}. \end{cases}$$

Indeed, in the equivalent case every intertwiner becomes a scalar multiple of a fixed equivalence isomorphism, while in the inequivalent case every intertwiner is zero.

Taking $E = D^{(\beta)}$ therefore gives

$$n_\beta = \dim_{\mathbb{C}} \text{Hom}_G(D^{(\beta)}, D). \quad (22.11)$$

The right-hand side depends only on the original representation D and the equivalence class of $D^{(\beta)}$, not on a chosen decomposition. Thus every multiplicity is intrinsic. Moreover, an irreducible class occurs in D exactly when this Hom space is nonzero, so the list of occurring classes is intrinsic as well. This proves uniqueness up to equivalence and reordering. ■

22.10 The defining representation of $GL(n, \mathbb{C})$

Example 22.2: Irreducibility of the defining representation

Let $G = GL(n, \mathbb{C})$ act on $V = \mathbb{C}^n$ by its defining representation

$$D : G \longrightarrow GL(\mathbb{C}^n), \quad D(A) = A.$$

We show that D is irreducible.

Suppose $U \leq \mathbb{C}^n$ is a nonzero proper invariant subspace. Let $\dim U = k$, where $0 < k < n$, choose a basis $(e_1, \dots, e_k)$ of U , and extend it to a basis $(e_1, \dots, e_n)$ of $\mathbb{C}^n$. Define an invertible linear map A by cyclically permuting this basis:

$$Ae_i = e_{i+1} \quad (1 \leq i < n), \quad Ae_n = e_1.$$

This A lies in $GL(n, \mathbb{C})$. But

$$e_k \in U, \quad Ae_k = e_{k+1} \notin U,$$

contradicting invariance under every element of $GL(n, \mathbb{C})$. Therefore the only invariant subspaces are $\{0\}$ and $\mathbb{C}^n$, so the defining representation is irreducible.

Example 22.3: The centre of $GL(n, \mathbb{C})$

Let $A \in GL(n, \mathbb{C})$ commute with every $B \in GL(n, \mathbb{C})$:

$$AB = BA \quad (B \in GL(n, \mathbb{C})).$$

Regard $B \mapsto B$ as the defining irreducible representation. The commutation relation says that A is an intertwiner from this representation to itself:

$$\begin{array}{ccc} \mathbb{C}^n & \xrightarrow{A} & \mathbb{C}^n \\ \downarrow B & & \downarrow B \\ \mathbb{C}^n & \xrightarrow{A} & \mathbb{C}^n \end{array}$$

By the scalar-commutant corollary to Schur's lemma,

$$A = \lambda I_n$$

for some $\lambda \in \mathbb{C}$. Since A is invertible, $\lambda \neq 0$. Conversely every nonzero scalar matrix commutes with every matrix. Hence

$$Z(GL(n, \mathbb{C})) = \{\lambda I_n : \lambda \in \mathbb{C}^\times\}.$$

What to retain from this chapter

1. $U \leq V$ is invariant if $D(a)U \subseteq U$ for every a . Restricting D to U gives a smaller representation D_U .
2. A representation is reducible exactly when it has a nontrivial invariant subspace. In an adapted basis all matrices have the simultaneous block form

$$D(a) = \begin{pmatrix} D_U(a) & N(a) \\ 0 & D''(a) \end{pmatrix}.$$

3. Complete reducibility requires invariant complementary subspaces and produces simultaneous block-diagonal form with irreducible blocks. Reducibility alone does not guarantee this.
4. If D is unitary and U is invariant, then $U^\perp$ is invariant. Repeated orthogonal splitting proves that every finite-dimensional unitary representation is completely reducible.
5. Maschke's theorem follows by averaging an inner product or a projection. A finite-group representation is completely reducible when the field characteristic does not divide $|G|$. The same is true for continuous finite-dimensional representations of compact groups.
6. Schur's lemma says that an intertwiner between irreducibles is either zero or an isomorphism. Over $\mathbb{C}$, every commuting endomorphism of a finite-dimensional irreducible representation is λI .
7. Finite-dimensional complex irreducible representations of Abelian groups are one-dimensional.
8. In a completely reducible complex representation, the irreducible equivalence classes and their multiplicities are unique. The multiplicity of E is

$$\dim_{\mathbb{C}} \operatorname{Hom}_G(E, D).$$